

Your grade: 100%

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Next item →

Instructions

1.

In the k -fold cross-validation technique, the data are shuffled randomly and then divided into k equal sub-samples. Which of the following statement(s) is/are true?

1 / 1 point

(I) We then use all the samples as training samples to be learned from.

(II) We then use $k - 1$ samples as training samples and the k -th sample as a test sample.

(III) We then use $k - 2$ samples as training samples, and 2 samples as test samples.

☐ I only

☒ II only

☐ III only

☐ II and III only

☐ I, II, and III

✔ Correct
2.

Which of the following R -command is/are related to setting up a robust regression using the MASS library?

1 / 1 point

(I) `rlm(y ~ x, data = df, psi = psi.bisquare)`

(II) `rlm(y ~ x1 + x2, data = df, psi = psi.huber)`

(III) `glm(y ~ x, data = df, family = binomial)`

☐ I only

☐ II only

☐ III only

☒ I and II only

☐ I, II, and III

✔ Correct
3.

Which of the following can be used to measure how good a model can predict the actual observations.

1 / 1 point

(I) A high R^2 score.

(II) A high MAE score.

(III) A high RMSE score.

☒ I only

☐ II only

☐ III only

☐ II and III only

☐ I, II, and III

✔ Correct
4.

Which of the following is an advantage of using the M -estimation technique?

1 / 1 point

☐ To detect and remediate multicollinearity issue in the data.

☐ To obtain estimations of the regression parameters using the least square method.

☐ To detect and delete outliers in the data.

☐ M -estimation is used as a variance stabilizing technique.

☒ M -estimation is a robust regression technique used when the data has outliers.

✔ Correct
5.

Which of the following can be used in the variable selection process?

1 / 1 point

(I) Ridge regression

(II) Forward stepwise regression

(III) Backward stepwise regression

☐ I only

☐ II only

☐ III only

☒ II and III only

☐ I, II, and III

✔ Correct